

$$\mathbb{N} = \{1, 2, 3, \dots\}$$

$$\mathbb{Z} = \{\dots, -3, -2, -1, 0, 1, 2, 3, \dots\}$$

~~$$\mathbb{Q} = \{ \}$$~~

$$\mathbb{R} = \{\dots, -3, -2.9, -2.8, -2, -1.5, -1, 0, 1, 1.5, \dots, 2, \dots, 3, \pi, \dots\}$$

$$\mathbb{N} \subseteq \mathbb{Z} \subseteq \mathbb{Q} \subseteq \mathbb{R}$$

$$\rightarrow \underbrace{\mathbb{N} \times \mathbb{N}}_{\text{grid of points}} \subseteq \underbrace{\mathbb{R} \times \mathbb{R}}_{\text{cartesian plane}}, \quad \text{why?}$$

$$\begin{array}{|l} \rightarrow \underline{A \subseteq A} \quad \checkmark \\ \rightarrow \underline{\emptyset \subseteq A} \quad \checkmark, \text{ why?} \end{array}$$

$$\text{ex: } A = \{1, 2, 3\}$$

$$\{\}, \{1\}, \{2\}, \{3\}, \{1, 2\}, \{1, 3\}, \{2, 3\}, \{1, 2, 3\}$$

$\rightarrow$ Proof $\emptyset \subseteq A$ for any set A .

Proof by Contradiction:

$\rightarrow$ we assume $\emptyset \not\subseteq A$.

$\rightarrow$ this means at least one element from empty set ($\emptyset$) is not an element in A .

$\rightarrow$ since empty set has no elements, the previous statement is contradicted.

$\rightarrow$ we arrive at a contradiction, thus proof is finished.

thus $\emptyset \subseteq A$ holds

thus $\emptyset \not\subseteq A$ does not hold.

$\rightarrow$ number of subsets of A

$2^{|A|}$

$\rightarrow$ proper subset